

# SIH26138: Executive Green Fleet Modernization & Decision Report

Maritime Decarbonization Intelligence | Generated: 2026-09-26 18:48:01 UTC

## 1. Executive Brief & Strategic Mandate

The proposed green fleet strategy transitions 1 vessels on the 3,500 nm corridor, achieving \$2.02M in annual operational savings (20.5%) and eliminating 4,822 metric tons of Well-to-Wake CO<sub>2</sub>e emissions (16.1%). Savings are decomposed as: fuel bunker differential \$1.19M [MODELLED], carbon cost savings \$0.39M [MODELLED], and FuelEU penalty avoidance \$0.44M [MODELLED]. With a retrofit CAPEX of \$19.5M [MODELLED] across 0 assets, the plan delivers a projected 10-year ROI of 3.51% with a simple payback of 9.7 years (EXTENDED). Crucially, schedule reliability is maintained at 100.0% through intelligent hydrodynamic eco-speed management (13.5 knots), ensuring statutory compliance under both IMO MEPC.400(83) and EU FuelEU Maritime regulations.

## 2. Financial & Modernization Investment Economics

| Metric                          | Value            | Operational Rationale                        |
|---------------------------------|------------------|----------------------------------------------|
| Total Retrofit Capex            | \$19.50M         | Shipyard dual-fuel conversion allocation     |
| Annual Net Benefit              | \$2.02M          | Net operational fuel and carbon tax savings  |
| 10-Year ROI                     | 3.51%            | Projected capital return over baseline       |
| Payback Period                  | 9.66 yrs         | Status: EXTENDED                             |
| Annual Cost Savings             | \$2.02M (20.47%) | Combined opex & regulatory penalty reduction |
| WTW CO <sub>2</sub> e Reduction | 4,822 t (16.09%) | Lifecycle decarbonization impact             |

## 3. Operational Risk Mitigation Matrix

| Operational Risk                                   | Strategic Impact                                                     | Mitigation Strategy                                                                               |
|----------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Alternative Green Fuel Price Spikes                | High opex inflation if green fuel market spread widens above \$400/t | Dual-fuel propulsion flexibility allows tactical switching back to MGO blended with green gasoil. |
| Shipyard Retrofit Bottlenecks & Delays             | Missed charter revenues and drydock overruns during engine overhaul  | Staggered retrofit schedule to convert no more than 20% of fleet concurrently.                    |
| Regulatory Tightening (IMO MEPC.400(83) Phase 1/2) | Operational downgrade to D/E resulting in mandated speed reductions  | Dynamic speed planning and wind-assist propulsion retrofit readiness.                             |

## 4. Evidence vs. Assumption Classification

| Item                                      | Category  | Regulatory / Physical Basis                                                                                                                  |
|-------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| IMO CII Reduction Targets (2026–2030)     | STATUTORY | Adopted under IMO Resolution MEPC.400(83) (Z = 11.0% to 21.5% relative to 2019 baseline).                                                    |
| EU FuelEU Maritime Milestones (2025–2040) | STATUTORY | Legally enacted under Regulation (EU) 2023/1805 (2025: −2%, 2030: −6%, 2035: −14.5%, 2040: −31% relative to 2020 baseline).                  |
| Annual Fuel Savings                       | MODELLED  | Bunker cost differential: Σ(m <sub>f</sub> × price <sub>f</sub> ) baseline – optimised. Fuel masses from fleet composition optimizer.        |
| Annual Carbon Cost Savings                | MODELLED  | ΔWTW CO <sub>2</sub> e (t) × ETS carbon price (\$/t). Emission values from fleet optimizer's Emission Engine using 2025-2040 FuelEU factors. |